AI-Powered Road Accountability Platform

RoadWatch is a mobile-first AI chatbot and map dashboard that consolidates road construction data, maintenance records, contractor details, and budget information into a single accessible platform — enabling citizens to monitor road quality, file complaints, and hold authorities accountable.

01 TAP-A-ROAD INTERFACE

A map-based interface where citizens tap any road to instantly view all associated metadata pulled from government APIs and OpenStreetMap — no searching required.

- Road Type displayed: NH (National Highway) / SH (State Highway) / MDR (Major District Road)
- Last relaying date and responsible contractor name shown on tap
- Budget sanctioned vs actually spent — visualised as a comparison bar
- Works offline for district-level cached data

⌖ Directly addresses: Data accuracy + Budget transparency evaluation criteria

02 AI-POWERED COMPLAINT PORTAL

Citizens upload a photo of a damaged road. The AI vision model analyses the image and auto-generates a structured complaint with severity classification and response deadlines.

- AI detects issue type: Pothole / Crack / Waterlogging / Missing Signage / Road Cave-in
- Assigns Severity Level — Low / Medium / High / Critical based on image analysis
- Auto-fills complaint form: GPS location, road name, road type, date
- Generates estimated response deadline based on severity

SEVERITY	RESPONSE SLA	ESCALATION
Critical	48 Hours	Immediate — Senior Authority
High	7 Days	Auto-escalates if unresolved
Medium	15 Days	Standard routing
Low	30 Days	Queued for scheduled repair

- If 3+ complaints filed on same road stretch → Auto-escalated to senior authority

⌖ Directly addresses: Complaint routing mechanism + UI/Accessibility criteria

03 SMART COMPLAINT ROUTING ENGINE

Complaints are routed automatically based on road type and jurisdiction boundaries — ensuring the right authority receives it every time, with no manual sorting.

ROAD TYPE	ROUTED TO	ESCALATION PATH
NH – National Highway	NHAI Regional Office	NHAI HQ → Ministry of Road Transport
SH – State Highway	State PWD Executive Engineer (Zone)	PWD Chief Engineer → State Govt
MDR / Village Road	District Collector / Panchayat Eng.	District Magistrate → State Rural Dept

- ■ Each authority mapped to a geographic jurisdiction boundary in the database
- ■ Complaint gets a unique tracking ID — citizen notified via SMS/app push
- ■ If no action taken within SLA → system auto-escalates one level up the hierarchy

✦ Directly addresses: Complaint routing mechanism + Accountability criteria

ACCOUNTABILITY SCORING SYSTEM

Every road and every contractor receives a public score from 0–100, calculated algorithmically from real data. Citizens and auditors can see these scores at any time.

ROAD HEALTH SCORE (0–100)

PARAMETER	WEIGHTAGE	HOW IT'S MEASURED
Days since last repair	25%	Current date minus last repair date
Open complaints count	25%	Active unresolved complaints on this road
Budget utilisation	20%	Amount spent vs sanctioned budget
Repair durability	30%	Re-complaints on same spot after repair

CONTRACTOR PERFORMANCE SCORE (0–100)

- ■ Aggregated score across ALL roads handled by the contractor
- ■ Tracks repeat failures — same road repaired 3× = red flag
- ■ Measures project completion: on time vs delayed vs abandoned
- ■ Public Contractor Leaderboard visible to all citizens and officials

■ RED FLAG SYSTEM: Any contractor scoring below 40 is automatically flagged publicly with the specific reason. Citizens can see this history before a new contract is awarded, enabling pre-emptive accountability.

05 REPAIR TIMELINE VIEW

Every road displays a complete chronological history of its life — from construction to every repair and complaint. This makes corruption patterns instantly visible.

Built 2018 → Repaired 2020 → Complaint 2022 → Repaired 2022 → ■ **Complaint 2024**

- ■ Frequently repaired roads are flagged — could indicate poor construction quality or inflated contracts
- ■ Timeline is publicly accessible; journalists and auditors can use it as evidence

✦ *Directly addresses: Budget transparency + Accountability criteria*

06 3-LEVEL BUDGET TRAIL

For every road project, citizens can see exactly where public money went — from government sanction all the way to ground-level spending.

BUDGET LEVEL	WHAT IT SHOWS	FLAG TRIGGER
Amount Sanctioned	Total funds approved by govt body	—
Amount Released	Funds actually transferred to contractor	If released > sanctioned
Amount Spent	Funds used on ground-level work	If large gap vs released

- ■ Any large gap between sanctioned and spent amount triggers an automatic audit flag
- ■ Unspent balances tracked — shows if funds were returned or diverted

✦ *Directly addresses: Budget transparency including the source — evaluation criteria*

07 COMMUNITY UPVOTING SYSTEM

Citizens can upvote existing complaints instead of filing duplicates. This creates a democratic priority system — the more people affected, the higher the urgency.

- ■ Each complaint shows upvote count — visible to authorities
- ■ High upvote complaints rise to top of the authority's action queue
- ■ Prevents database clutter from duplicate complaints on the same road
- ■ Citizens receive update notifications when a complaint they upvoted gets resolved

WHISTLEBLOWER MODE

An anonymous reporting channel for citizens who want to report contractor malpractice, corruption, or deliberate negligence without fear of identification or retaliation.

- ■ Reports submitted anonymously — no personal data stored or linked
- ■ Bypass normal authority routing — reports go directly to a third-party audit body
- ■ Citizen can attach images, documents, or voice notes as evidence
- ■ Report status trackable via anonymous ID code (no login required)

> ★ *STANDOUT IDEA: This feature differentiates RoadWatch from standard complaint portals. By providing a safe, anonymous channel backed by a third-party auditor, it enables real whistleblowing — not just surface-level complaints.*

09 CITY / DISTRICT PUBLIC DASHBOARD

A public-facing web dashboard showing the overall road health of a city or district at a glance — useful for citizens, journalists, policymakers, and NGOs.

METRIC	EXAMPLE
Total roads in district	1,247 roads mapped
Road condition split	42% Good · 35% Average · 23% Poor
Total complaints (month)	318 filed · 204 resolved · 114 pending
Average resolution time	11.4 days (target: 7 days)
Lowest scoring contractor	Flagged publicly with score history

- ■ Dashboard data updates in real-time as complaints are filed and resolved
- ■ Exportable as PDF/CSV for use in RTI applications or news reporting

10 OFFLINE MODE + GLOBAL APPLICABILITY

Built for low-connectivity environments and designed to work across any country with publicly available road and government data.

- ■ District-level road data cached locally on device — works fully offline
- ■ Complaints saved locally and auto-synced when connectivity is restored
- ■ Data ingested from OpenStreetMap + government open data portals
- ■ Country-agnostic: works for India, Kenya, Brazil — any country with public road APIs
- ■ Regional language support via AI translation layer

⌖ *Directly addresses: Offline functionality + Global applicability + UI/Accessibility criteria*

EVALUATION CRITERIA COVERAGE

CRITERIA	HOW ROADWATCH ADDRESSES IT	FEATURE
Data Accuracy	Live govt API + OSM + AI image verification	01, 02
Complaint Routing	Hierarchy-based jurisdiction mapping with SLA escalation	03
Budget Transparency	3-level budget trail with auto-flagging of gaps	06
UI & Accessibility	Image upload, voice input, regional language, offline mode	02, 10
Global Integration	Country-agnostic data layer via OpenStreetMap	10
Accountability	Dual scoring (road + contractor) with public leaderboard	04

RECOMMENDED TECH STACK

Chosen for intermediate-level developers — all tools are open-source, well-documented, and deployable without paid infrastructure. Web app architecture (not mobile).

LAYER	PRIMARY CHOICE	ALTERNATIVE	PURPOSE
Frontend UI	**React + Tailwind CSS**	Next.js + Tailwind	Web app interface, responsive design
Map & Roads	**Leaflet.js + OpenStreetMap**	Mapbox GL JS (free tier)	Tap-a-road interface, boundary mapping
Backend API	**FastAPI (Python)**	Express.js (Node)	REST APIs, routing logic, scoring engine
Authentication	**JWT + bcrypt**	Flask-Login / Passport.js	Stateless auth, secure token-based sessions
Database	**PostgreSQL + SQLAlchemy**	MongoDB + Mongoose	Road data, complaints, user records
Image Analysis (AI/ML)	**YOLOv8 (Ultralytics)**	Custom CNN (TensorFlow/Keras)	Pothole/crack detection, severity scoring
ML Training	**PyTorch + Roboflow dataset**	TensorFlow + Kaggle dataset	Train on labelled road damage images
Image Storage	**Cloudinary (free tier)**	AWS S3 / Firebase Storage	Store complaint photos uploaded by citizens
Notifications	**Firebase Cloud Messaging**	OneSignal (free tier)	Complaint status updates, escalation alerts
Deployment	**Render (backend) + Vercel (frontend)**	Railway / Fly.io	Free-tier cloud hosting, no credit card needed

Offline Support	**Service Workers + IndexedDB**	PWA (Progressive Web App)	Cache district data, queue offline complaints

■ ML IMAGE ANALYSIS DETAIL: YOLOv8 is recommended as primary because it is fast, pre-trainable on road damage datasets from Roboflow, and runs inference in under 200ms. As an alternative, a custom CNN built with TensorFlow/Keras can be trained on the IEEE Road Damage Dataset or Kaggle pothole datasets — suitable for intermediate ML developers. Both approaches output: damage class + confidence score → mapped to severity level.

■ AUTH DETAIL: JWT (JSON Web Tokens) with bcrypt password hashing handles all authentication stateless — no session storage needed. Access tokens (15 min expiry) and refresh tokens (7 days) issued on login. Role-based access: Citizen / Authority / Admin. Alternatives: Flask-Login for simpler session-based auth, or Passport.js if using Node backend.